

Week 6 Discussion: Forwards, Futures & Swaps (Lec 14–16)

Main focus: (i) Pricing vs. valuing a forward; (ii) futures pricing with income/costs (Master Formula); (iii) designing an interest rate swap via comparative advantage.

1 Part 1: Value of a Forward Contract

Module A) Pricing (F_0) vs. Valuation (f_t)

Crucial Distinction:

- **Forward Price (F_0):** the delivery price K is set at inception so that the contract value is 0.
- **Value (f_t):** the mark-to-market value of an *existing* contract at time t .

Non-dividend-paying stock (continuous compounding):

$$F_0 = S_0 e^{rT}.$$

Value of an existing long forward at time t :

$$f_t = S_t - K e^{-r(T-t)} \quad \text{and} \quad f_t = (F_t - K) e^{-r(T-t)}, \quad F_t = S_t e^{r(T-t)}.$$

Practice A: (Textbook 5.7) Forward price and value

A 1-year long forward contract on a non-dividend-paying stock is entered into when the stock price is $S_0 = \$40$ and the risk-free rate is $r = 5\%$ per annum with continuous compounding.

- What are the **forward price** (F_0) and the **initial value** of the forward contract?
- Six months later, the stock price is $S_{0.5} = \$45$ and the risk-free rate is still 5%. What are the **forward price** for delivery in 6 months and the **value** of the original forward contract?

2 Part 2: Pricing Forwards with Income / Costs / Currencies

Module B) The “Master Formula” (Cost of Carry)

The No-Arbitrage Forward Price F_0 relates to Spot Price S_0 via:

$$F_0 = S_0 e^{(r-q+u-y)T} \quad \text{or} \quad F_0 = (S_0 - I + U) e^{rT}$$

- **Interest (r):** Cost of financing. **Push F Up.**
- **Income (q or I):** Dividends/Coupons. Holder of forward *misses* these. **Push F Down.**
- **Storage (u or U):** Cost to hold physical asset. Holder of forward *avoids* these. **Push F Up.**
- **Convenience Yield (y):** Benefit of holding physical asset. **Push F Down.**

Special Case: Currencies (Interest Rate Parity) Treat foreign risk-free rate r_f as a dividend yield q .

$$F_0 = S_0 e^{(r-r_f)T} \quad \text{where } r = r_{\text{domestic}}, r_f = r_{\text{foreign}}.$$

Practice B: (Textbook 5.10) Index futures mispricing

Suppose the risk-free interest rate is 6% per annum with continuous compounding and the dividend yield

on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405.

- (a) What is the **theoretical** no-arbitrage futures price? (Hint: Index has yield q , no storage/convenience).
- (b) What arbitrage opportunity does this create? (State the trades at $t = 0$ and what happens at maturity.)

3 Part 3: Interest Rate Swaps

Module C) Designing Swaps (Comparative Advantage + Bank Spread)

Setup: Two firms have different borrowing costs in fixed vs floating markets. A swap can create a *total gain*:

$$\text{Gain} = (\text{Fixed spread}) - (\text{Floating spread}).$$

- The stronger-credit firm typically has a bigger advantage in the **fixed** market.
- If a bank intermediates, it takes a spread; the remainder is split between the two firms (often equally if stated).
- Design via **target effective rates** (what each firm ends up paying) and then back out the swap fixed rates.

Practice C: (Textbook 7.1) Swap design with bank spread

Companies A and B have been offered the following rates per annum on a \$20 million five-year loan:

	Fixed rate	Floating rate
Company A	5.0%	SOFR + 0.1%
Company B	6.4%	SOFR + 0.6%

Company A requires a floating-rate loan; Company B requires a fixed-rate loan. Design a swap that will net a bank (acting as intermediary) **0.1% per annum** and that will appear **equally attractive** to both companies.

- (a) Compute the **total gain** available before the bank spread.
- (b) If the bank takes 0.1% and the rest is split equally, what **effective** borrowing rates should A and B end up with?
- (c) Give one concrete swap with the bank: specify who borrows what externally and the swap cashflows (fixed rates paid/received, floating legs).